PassFrame
tuition

Coordinates & Transformations

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: A point is at (-9, 6). It is rotated 90° clockwise about the origin using (x, y) → (y, -x), and the result is then translated 4 left. What are the final coordinates?

Method & answer: Rotate: (y, -x) = (6, 9). Translate 4 left: (2, 9).

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point A is at (-14, 17). What is its x-coordinate?

2. A point A is at (-14, 17). What is its y-coordinate?

3. A point starts at (-18, 9) and is translated 22 right and 27 down. What are its new coordinates?

4. A point starts at (13, -19) and is translated 28 left and 24 up. What are its new coordinates?

5. A point is at (-21, 12). It is translated 25 right. What are its new coordinates? A) (4, 12) B) (-46, 12) C) (-21, 37) D) (-21, -13)

6. A point is at (-16, -13). Which quadrant of the grid is it in?

7. A point is at (19, -9). Which quadrant of the grid is it in?

8. A point starts at (-22, 8) and is translated 13 right and 17 down. What are its new coordinates?

9. A point starts at (11, 24) and is translated 31 left and 13 down. What are its new coordinates?

10. Select ALL that apply. Look at the points P(-8, 19), Q(8, -19), R(-8, -19) and S(8, 19). Which do NOT lie in the third quadrant? A) P(-8, 19) B) Q(8, -19) C) R(-8, -19) D) S(8, 19)

11. A point starts at (0, -23) and is translated 17 left and 27 up. What are its new coordinates?

12. A point starts at (-25, -18) and is translated 34 right and 29 up. What are its new coordinates? A) (9, 11) B) (-59, -47) C) (9, -47) D) (-59, 11)

13. A line segment joins A(-12, 5) and B(14, 5). What is the midpoint of AB?

14. A point starts at (17, -11) and is translated 27 left and 19 up. What are its new coordinates?

15. Which of these points lies exactly on the y-axis? A) (0, -17) B) (-17, 0) C) (17, 17) D) (-17, -17)

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point is at (-19, 12). It is reflected in the x-axis. What are its new coordinates?

2. A point is at (21, -15). It is reflected in the y-axis. What are its new coordinates?

6. A point is at (16, 11). It is reflected in the x-axis, then the result is reflected in the y-axis. What are the final coordinates?

7. A point is at (-24, -19). It is reflected in the y-axis. What are its new coordinates?

11. A point is at (-27, -20). It is reflected in the x-axis. What are its new coordinates?

12. Select ALL that apply. A rectangle has vertices at (-11, -5), (16, -5), (16, 12) and (-11, 12). Which statements are true? A) Its width is 27 units B) Its height is 17 units C) The vertex (16, 12) is diagonally opposite (-11, -5)

3. A point starts at (-23, 18) and is translated 28 right and 22 down. What are its new coordinates?

4. A shape has vertices A(-8, -8), B(-8, 3), C(4, 3) and D(4, -8). It is translated 10 right and 10 up. What are the new coordinates of A, B, C and D?

5. A point at (-17, -9) is reflected in the x-axis. What are its new coordinates? A) (-17, 9) B) (17, -9) C) (17, 9) D) (-17, -9)

8. A line segment joins A(-10, -7) and B(-10, 17). What is the length of AB, and what is its midpoint?

9. A point starts at (26, -21) and is translated 35 left and 15 up. What are its new coordinates?

10. A point at (22, -15) is reflected in the y-axis. What are its new coordinates? A) (-22, -15) B) (22, 15) C) (-22, 15) D) (15, -22)

13. A point starts at (-5, -5) and is translated 30 right and 30 up. What are its new coordinates?

14. A point at (-6, 18) is reflected in the x-axis to give P, then P is translated 9 down. What are the final coordinates?

15. A point starts at (-16, 19) and is translated 24 right and 27 down. What are its new coordinates? A) (8, -8) B) (-40, -8) C) (8, 46) D) (-40, 46)

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point at (4, -9) is rotated 90° clockwise about the origin using (x, y) → (y, -x), then the result is translated 3 left. What are the final coordinates?

2. A rectangle has vertices at (-8, -5), (12, -5), (12, 9) and (-8, 9). Is its perimeter greater than or less than 80 units?

3. A point is at (-14, -9). It is reflected in the y-axis to give point P. What is the sum of the x-coordinate and y-coordinate of P?

4. A point at (17, -6) is translated 17 left, landing exactly on which axis?

5. A square has vertices at (-6, -6), (10, -6), (10, 10) and (-6, 10). What is the coordinate of its centre, and is the centre in the first quadrant?

6. A point is at (-9, 16). It is rotated 90° clockwise about the origin. What are its new coordinates?

7. A rectangle has vertices at (-12, -4), (8, -4), (8, 6) and (-12, 6). What is its perimeter, and what are the coordinates of its centre?

8. A point is at (-11, -4). It is rotated 90° anticlockwise about the origin. What are its new coordinates?

9. A point is at (14, -17). It is rotated 180° about the origin. What are its new coordinates?

10. A point is at (-21, 19). It is reflected in the y-axis. What are its new coordinates?

11. Select ALL that apply. A shape has vertices A(-4, 0), B(-4, 11), C(7, 11) and D(7, 0). It is rotated 180° about the origin. Which of these are correct new vertices? A) A becomes (0, -4) B) B becomes (4, -11) C) C becomes (-7, -11) D) D becomes (-7, 0)

12. A point starts at (19, -24) and is translated 33 left and 35 up. What are its new coordinates?

13. A point at (13, -19) is reflected in the y-axis, and the result is then rotated 90° clockwise about the origin using (x, y) → (y, -x). What are the final coordinates?

14. A rectangle has vertices at (-14, -10), (6, -10), (6, 8) and (-14, 8). What is its perimeter? A) 76 units B) 38 units C) 58 units D) 96 units

15. A triangle has vertices at (3, 2), (3, 9) and (8, 2). It is translated 5 left and 4 down. What are the new coordinates of all three vertices?